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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B. E. Fast Track Examinations Oct-2020

Electronics and Communication Engineering

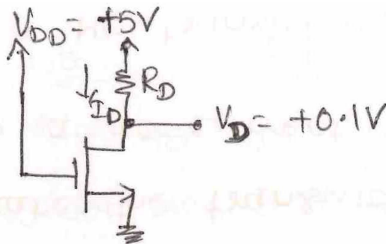
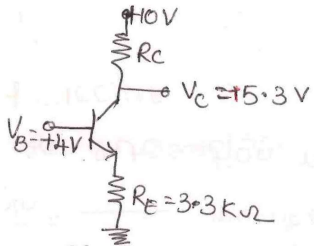
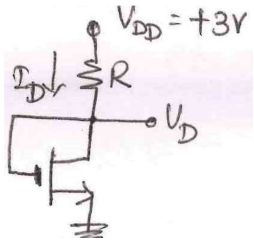
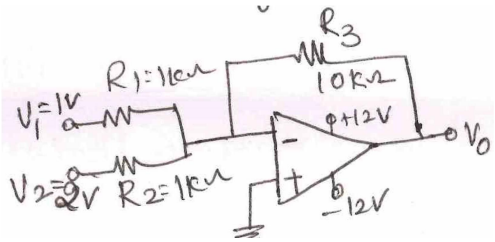
ANALOG MICROELECTRONIC CIRCUITS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

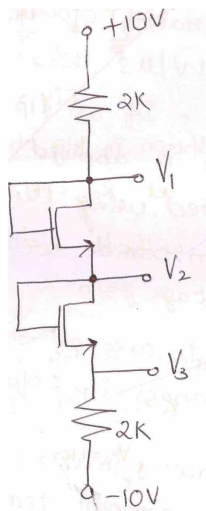
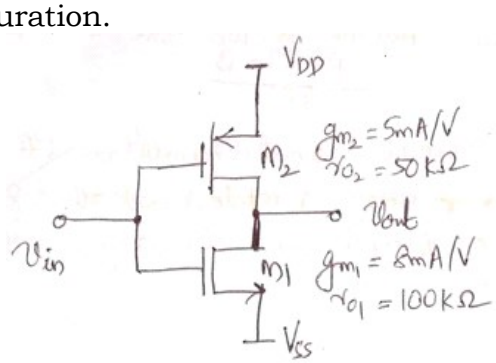
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

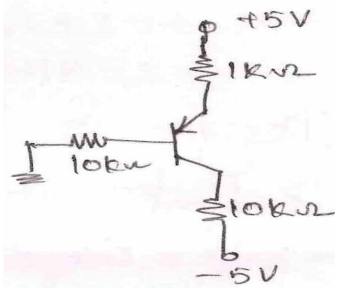
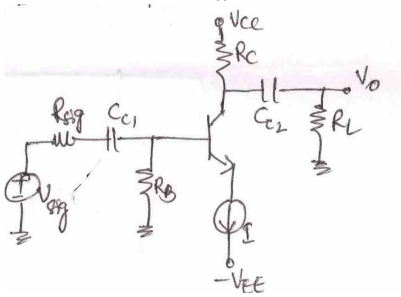
PART-A

1	1.1	For a $MOSFET$ with $w/L = 8\mu m/0.8\mu m$, $\mu_n C_{ox} = k_n' = 194\mu A/V^2$, what are the values of $V_{GS} =$ _____ & $V_{DSmin} =$ _____ needed to operate the transistor in saturation region with a DC current of $100\mu A$.	02
	1.2	In a MOS transistor, g_m is _____, if dc bias current increases by 4 times while maintaining other parameters constant.	01
	1.3	Given $k_n'\left(\frac{w}{L}\right) = 1mA/V^2$, $V_t = 1V$ & $V_D = 0.1V$ for an $NMOS$ transistor shown in the Fig. 1.3. Find R_D= _____.  Fig. 1.3  Fig. 1.4	02
	1.4	For the circuit shown in Fig. 1.4, find $R_c =$ _____ if $V_{CC} = 10V$, $\beta = 100$ & $V_{CE} = 2V$.	01
	1.5	A BJT having $\beta = 100$ is biased at a dc collector current of $1mA$. Find the value of g_m & r_e.	02
	1.6	Including R_e in emitter of CE amplifier will increase the input resistance by a factor of _____.	01
	1.7	Calculate f_T for the n-channel $MOSFET$ with $g_m = 1mA/V$, $C_{gs} = 60fF$ & $C_{gd} = 30fF$.	01
	1.8	In an $NMOS$ transistor with $I_D = 80\mu A$, $V_D = 1V$, find the value of R in the Fig. 1.8.  	

	1.9	For a transistor with $\beta = 100$ & $r_0 = 100k\Omega$. Find the output resistance of the Wilson mirror circuit.	01
	1.10	Mention any two main advantages of Darlington configuration.	02
	1.11	For the circuit shown in Fig. 1.11, find the output voltage.	02
	1.12	The negative feedback in the case of a voltage series feedback amplifier _____ input impedance by $1 + A\beta$ and _____ output impedance by $1 + A\beta$.	02
	1.13	A BJT is specified to have a maximum power dissipation of $2W$ at an ambient temperature of $25°C$, and a maximum junction temperature of $150°C$. Find the thermal resistance.	01
	1.14	A class B push pull amplifier with $V_{CC} = 50V$, $I_m = 4.3A$ & $V_{min} = 5V$. Find the conversion efficiency.	01

PART-B

2	a	Calculate the values of V_1, V_2 and V_3 of the circuit in Fig. 2.a, assuming the transistors are identical. Given $\mu_n C_{ox}(w/L) = 2mA/V^2$ and $V_t = 1V$.  Fig. 2.a	10
	b	Draw the small signal equivalent circuit of the $CMOS$ inverter shown in Fig. 2.b. Determine the small signal gain of the circuit assuming transistors in saturation.  Fig. 2.b	06
3	a	Draw the circuit diagram of a Darlington pair configuration and obtain the expression for input impedance and voltage gain.	10
	b	Design a Widlar current source to have a output current of $1\mu A$. Given $I_{REF} = 100\mu A$, $V_{DD} = 5V$. Assume $V_{BE} = 0.7V$ for $1mA$ and large β. **OR**	06

4	a	Draw the circuit of CE amplifier & its equivalent circuit using $hybrid-\pi$ model and then derive expressions for voltage gain, input and output impedance.	10
	b	For the circuit shown in Fig. 4.b, determine the voltages at all nodes and the current through all branches. Assume β as 30.  Fig. 4.b	06
5	a	For the circuit shown in Fig. 5.a, determine he midband gain and the upper $3dB$ frequency of CE amplifier. Given $V_{CC}=V_{EE}=10V$, $I=1mA, R_B=100k\Omega$, $R_C=8k\Omega$, $R_{sig}=5k\Omega$, $R_L=5k\Omega$, $\beta_0=100$, $V_A=100V$, $C_\mu=1pF$, $f_T=800MHz$ & $r_x=50\Omega$  Fig. 5.a	08
	b	Find the midband gain A_m and upper $3dB$ frequency f_H of a CS amplifier having $R_{sig}=100k\Omega$, $R_G=4.7M\Omega$, $R_D=R_L=15k\Omega$, $g_m=1mA/V$, $r_0=150k\Omega$, $C_{gs}=1pF$ & $C_{gd}=0.4pF$.	08
		OR	
6	a	Draw the circuit diagram of Wilson Current Mirror using bipolar transistor and prove that output impedance is approximately $\frac{\beta r_0}{2}$. Assume that all transistors are identical.	08
	b	Draw the circuit of a cascade MOS current mirror. Obtain the expression for O/P resistance and minimum dc voltage.	08
7	a	Draw the circuit of instrumentation amplifier and explain its working. Also derive the expressions for gain of the circuit.	10
	b	Draw the circuit of Wein bridge oscillator using op-amp and explain its working with necessary equation & waveforms.	06
8	a	Draw the circuit of Class B transformer coupled amplifier and derive an expression for conversion efficiency.	04
	b	A BJT has $T_{jmax}=150°C$ and maximum power dissipation as: $40W\ at\ T_c=25°C$ & $20W\ at\ T_A=25°C$. Above 25°C, P_{Dmax} derated linearly with $\theta_{jc}=3.12°C/W$ and $\theta_{jA}=62.5°C/W$. Find the following: i) P_{Dmax} when operated in free air at $T_A=50°C$. ii) P_{Dmax} when operated in ambient temperature of 50°C, but with a heat sink for which $\theta_{cs}=0.5°C/W$ & $\theta_{SA}=4°C/W$. Find the temperature of case and heat sink	04